ChatGPT

Fear of Temperature: Second-Round Deep Research and Feasibility Assessment

Executive feasibility verdict and research boundary

Overall verdict: MODIFY, then proceed. The thesis is viable as a historically comparative, metadata-led study of how warming becomes measurable, emotionally meaningful, publicly communicable, contestable, and preservable. It is **not viable as a comprehensive global history of "climate fear," nor as a project that presumes one worldwide starting date or a single continuous Global Fear Index**.

The strongest Master-level design is:

> **A global metadata backbone, combined with three deliberately contrasting regional or national deep cases, a small multilingual media corpus, and a bounded heritage-persistence registry.**

The proposed database should therefore be constructed as a **linked evidence registry**, not as a universal archive claiming exhaustive global coverage. Its core entities should be climate observations, scientific and institutional documents, survey questions, media records, cultural artefacts, and evidence of preservation or reuse. "Fear" should appear only where a source directly measures or clearly expresses fear, worry, anxiety, apprehension, dread, or closely defined affect.

The bracketed first-round materials—object definition, main research question, semantic taxonomy, eligibility matrix, inclusion rules, and ethics requirements—were not actually supplied in the prompt. Consequently, the present report treats the following as **provisional rather than inherited conclusions**:

> **Provisional one-sentence object definition:** A *Fear of Temperature* heritage object is a bounded, source-identifiable record that measures, represents, expresses, contests, preserves, or later reuses meanings attached to rising temperature or climatic warming, with documented provenance and a defensible classification of whether its evidence concerns affect, risk, salience, policy, behaviour, or heritage persistence.

A candidate object should pass six tests: it must be bounded; have recoverable provenance; be relevant to temperature or warming rather than environment in general; have a classifiable evidentiary function; have lawful and ethical conditions of use; and, where described as heritage, have evidence of preservation, recognition, participation, transmission, or later reuse. Mere online availability does not by itself constitute heritage value.

The central feasibility finding is that the project has four strong evidentiary pillars but not one seamless global series. Global and regional temperature reconstruction is strong from the nineteenth century onward; scientific and institutional documentation is exceptionally strong; public-opinion evidence becomes substantial from the late 1980s and cross-nationally structured from the 1990s; and systematic evidence of heritage-making is strongest from the 2000s onward. These pillars begin at different dates and answer different claims. HadCRUT5 begins in 1850, NASA GISTEMP in 1880, ERA5 in 1940, and individual station series can reach much further back, but these products differ in variable construction, spatial treatment, resolution, uncertainty, and anomaly baseline. [1]

The thesis should distinguish five kinds of “beginning”:

Beginning type	Meaning in this project	Appropriate example
First occurrence	Earliest located formulation or record, even if isolated	Fourier’s atmospheric heat argument in 1824; Arrhenius’s quantitative calculations in 1896
Institutional milestone	Formal incorporation into an authoritative organisation or legal process	IPCC establishment in 1988; UNFCCC adoption in 1992
Public breakthrough	Marked increase in attention in a specific media and political system	The 1988 Hansen hearing and associated US media attention
Globally comparable period	A period with sufficiently harmonised cross-national evidence	ISSP from 1993 for environmental-risk items; World Risk Poll from 2019 for broad threat perception
Analytical thesis boundary	A chosen boundary justified by source density and research question	Recommended core boundary: 1988–2026, with prehistory back to 1824

The word **global** should mean “globally extensible metadata architecture with explicitly measured coverage gaps,” not “equal source density in every country, language, or community.” Survey infrastructure, newspaper digitisation, stable web archiving, cataloguing, and public access are strongly unequal. GDELT monitors material in many languages but changes its source base and technical architecture over time; MeCCO currently monitors selected outlets in 59 countries and 14 languages, not the total media systems of those countries; and major cross-national surveys do not necessarily measure the same affective construct. [2]

Recommended main analytical window: 1988–2026.

Archive prehistory window: 1824–1987.

Quantitatively comparable windows: 1993–2020 for repeated ISSP environmental-risk items, with wording discontinuities; 2009 onward for recurring European climate-specific surveys; 2019 onward for near-global threat-perception data; and country-specific windows such as Gallup’s US worry series from 1989. [3]

Multi-clock starting points and historical periodisation

Multi-clock starting-point table

Clock	Earliest located evidence	Earliest defensible starting point	Uncertainty and geographic or language limits	Primary source	Secondary interpretation	Su st
Scientific conceptualisation	Fourier's 1824 discussion of atmospheric retention of heat	**1896** for a quantitative CO_2–temperature calculation; **1938** for an explicit fossil-fuel CO_2–observed warming argument	Fourier's formulation was not the fully developed modern greenhouse theory. Arrhenius quantified climatic effects but should not be simplified into the exact modern attribution argument. Callendar's 1938 claim was more explicitly anthropogenic but remained disputed. The evidence is overwhelmingly European and English/French-language in its surviving canonical form.	Fourier, Tyndall, Arrhenius, and Callendar papers. Tyndall experimentally examined infrared absorption by gases; Arrhenius calculated temperature responses; Callendar connected fuel combustion, rising CO_2, and observed warming. [4]	Histories of climate science treat these as distinct stages rather than one discovery moment. [5]	**N** m bo 18 in pr 19 ar at m

Clock	Earliest located evidence	Earliest defensible starting point	Uncertainty and geographic or language limits	Primary source	Secondary interpretation	Su st
Instrumental and data	Long national or local instrumental series, including observations well before a global network	**1850** for a reconstructed global surface-temperature backbone using HadCRUT5; **1880** as an alternative for GISTEMP; **1940** for global hourly reanalysis with ERA5	Pre-twentieth-century spatial coverage is sparse and uneven, especially over oceans, polar regions, Africa, parts of Asia, and the Southern Hemisphere. Reanalysis is model-assimilated, not simply observed temperature. Station series may be long but are not automatically globally representative.	HadCRUT5, GISTEMP, ERA5, GHCN-D, Berkeley Earth and national meteorological archives. [6]	Dataset documentation emphasises changing coverage, homogenisation, reconstruction, and uncertainty. ECMWF specifically cautions that observing-system quantity and quality change over time. [7]	**Ye pl ba fo af** 18 lo co EF da ar
Institutional recognition	Government-commissioned scientific assessments and international climate conferences in the 1970s	**1988** for permanent international assessment through the IPCC; **1992** for treaty-level recognition through the UNFCCC	Earlier institutional concern existed, but the institutional field was fragmented. IPCC establishment does not mean equivalent national recognition or public acceptance everywhere. Treaty participation does not demonstrate public fear.	IPCC was established by WMO and UNEP in 1988; the UNFCCC was adopted on 9 May 1992. [8]	The IPCC assessment cycle consolidated scientific knowledge into an enduring international assessment infrastructure that informed subsequent negotiations.	**Ye** ex in ar bo pr no as bi fe

Clock	Earliest located evidence	Earliest defensible starting point	Uncertainty and geographic or language limits	Primary source	Secondary interpretation	Su st
Mass-media and public salience	Wider public discussion began emerging in the 1970s, while isolated earlier reporting also exists	**1988** for a breakthrough in the United States and parts of Europe; **2000–2004** for a more defensible multinational media-monitoring window	The 1988 breakthrough was strongly shaped by US heat, congressional testimony, political attention, and elite media. Comparable timing differed across national systems. Systematic multinational monitoring is much later and remains outlet-selective.	Hansen's 23 June 1988 Senate testimony; national newspaper and broadcast archives. [9]	Media histories identify 1988 as a major shift in several countries, but comparative work demonstrates different drivers and trajectories in Australia, Germany, India, the UK, and elsewhere. [10]	**Ye se co br nc ur da** or m re m m cc th M
Direct affect measurement	A 1988 European Community survey is reported as asking whether respondents were worried about the greenhouse effect; early US polls in 1986 asked whether warming was a serious problem	**1989** for a defensible, documented US worry series; **1988 provisionally** for regional European direct worry pending verification of the original questionnaire and translations	"Serious problem" is not identical to worry or fear. The 1988 EC result is known through scholarly synthesis, but the exact primary questionnaire, translations, sampling documentation, and variable history must be retrieved before treating it as the earliest anchor.	Gallup asks how much respondents personally worry about global warming; its trend begins in 1989. Early US polling in July 1986 asked about seriousness, not necessarily affect. [11]	A review of European and US public views reports that 76% across the then 12 EC member states were very or somewhat worried in 1988, but this remains a regional, not global, indicator. [12]	**19 de or di se** a

Clock	Earliest located evidence	Earliest defensible starting point	Uncertainty and geographic or language limits	Primary source	Secondary interpretation	Su st
Cross-national comparability	Regional Eurobarometer items around 1988–1989; the 1992 Health of the Planet survey covered 24 countries	**1993** for an archived harmonised multinational environment module; **2000** for a broad 34-country seriousness item; **2019** for much wider global threat-perception coverage	ISSP is not a pure climate-fear survey and changes terminology from "greenhouse effect/global warming" to "climate change." The 1992 Health of the Planet materials require questionnaire and microdata verification. World Risk Poll measures threat, not fear.	ISSP Environment 1993; GlobeScan 2000 seriousness question; World Risk Poll 2019, 2021 and 2023 data. [13]	The ISSP waves are partial replications, and source questionnaires explicitly document terminology changes, limiting simple trend claims. [14]	**Ye fo co bu th** N th co m de fr
Heritage persistence	Retrospective republication and canonisation of nineteenth- and early-twentieth-century papers; preservation of scientific records in specialist archives	**2005–2006** for formal archival preservation of a major climate-science record; **2019–2020** for active museum collecting of movement artefacts; **2025** for formal international documentary-heritage recognition of meteorological cooperation records	A source's age or fame is not itself heritage-making. Formal accession, community custody, exhibition, educational reuse, anniversary commemoration, or registry inscription must be demonstrated. Existing examples are institutionally and Anglophone-biased.	Charles David Keeling papers; Bristol Museums climate-movement collecting; Museums Victoria Mallee oral histories; WMO's IMO Legacy Collection inscription in UNESCO's Memory of the World Register. [15]	Documentary-heritage frameworks emphasise preservation, access, significance, and public awareness rather than mere survival. [16]	**N da** se sc ar m he co m fo re su

The most important methodological result is therefore:

> **1824, 1896, 1938, 1850, 1979, 1988, 1989, 1992, 1993, 2000, 2019, 2020, and 2025 can all be valid starting points—but for different claims.**

Working historical periodisation

Period	Beginning and ending rationale	Physical and scientific context	Institutional, media and affect context	Representative terminology and artefacts	Data and bias conditions
Conceptual and instrumental prehistory, 1824–1937	Begins with early physical reasoning about atmospheric heat and ends before Callendar's anthropogenic-observational synthesis	Atmospheric heat retention, gas absorption, early CO_2 calculations; expanding but highly uneven instrumental observation	Climate change was primarily a scientific and technical matter, not a continuously measured public affect	"Temperature of the globe," "carbonic acid," atmospheric absorption; Fourier, Tyndall, Arrhenius texts	Excellent canonical texts, weak public-opinion evidence; European scientific-language dominance
Anthropogenic proposition and monitoring formation, 1938–1978	Starts with Callendar's explicit fossil-fuel warming argument and ends before the Charney-era assessment consolidation	Growing CO_2 measurement, computer modelling, global monitoring, post-war earth-system science	Intermittent governmental and media attention; wider environmental anxiety by the 1960s–1970s, but little direct climate-affect polling	"CO_2 theory," "greenhouse effect," "climatic change"; Callendar paper, Keeling Curve, early models	Strong scientific archives; scattered media records; virtually no harmonised direct-affect data
Assessment emergence and early public recognition, 1979–1987	Begins with high-level scientific assessment and ends immediately before the 1988 institutional-media conjunction	Model-based estimates of warming and climate sensitivity become policy-relevant	Early governmental reports, hearings and news coverage; US awareness polling and seriousness questions appear	"Carbon dioxide and climate," "global warming," "greenhouse effect"	National and Anglophone source bias; polling questions are sparse and unstable

Period	Beginning and ending rationale	Physical and scientific context	Institutional, media and affect context	Representative terminology and artefacts	Data and bias conditions
Breakthroughs without simultaneity, 1988–1994	Begins with IPCC creation, Hansen's hearing and heightened media attention; ends after the first IPCC report, UNFCCC adoption and first ISSP Environment wave	Observational warming and model projections become more visible and institutionally coordinated	Public breakthroughs occur at different times in different countries; first defensible worry and regional comparison evidence; treaty institutionalisation	"Global warming," "greenhouse effect," "global climate change"; hearings, newspaper front pages, early polling, UNFCCC	Rich US and European evidence; weak basis for claims about a uniform global breakthrough
Treaty consolidation and comparative survey growth, 1995–2006	From post-UNFCCC implementation through Kyoto and the expansion of cross-national environmental polling, ending before the major 2007 visibility surge	Stronger attribution evidence, expanding observational datasets, increasingly mature modelling	Kyoto negotiations; ISSP 2000; GlobeScan comparisons; growing documentary and broadcast production	"Dangerous interference," "emissions targets," "climate risk," "2°C"	Better cross-national data, but survey wording and country participation remain discontinuous
Threshold, digital-media and polarisation phase, 2007–2014	Begins with intensified assessment, documentary and digital visibility and ends before Paris and the IPCC 1.5°C mandate	Attribution and observed impacts become more robust; regional and extreme-event analysis expands	High-profile documentaries, online news, sceptical countermobilisation, recurring climate surveys and searchable digital media	"Tipping point," "climate crisis," "2°C," adaptation, denial; documentary films and web-native campaigns	Searchable media grows rapidly, but platform and digitisation bias complicate longitudinal comparison

Period	Beginning and ending rationale	Physical and scientific context	Institutional, media and affect context	Representative terminology and artefacts	Data and bias conditions
Emergency, 1.5°C, anxiety and heritage-making, 2015–2026	Begins with the Paris Agreement and continues through climate-emergency declarations, youth mobilisation, validated anxiety scales, active collecting and formal heritage recognition	Record warmth and event attribution increase public relevance; 1.5°C becomes a central institutional threshold	Climate anxiety and worry scales proliferate; youth movements and emergency language spread; museums and archives actively collect climate materials	"1.5°C," "climate emergency," "climate anxiety," "loss and damage," "tipping points"; protest placards, oral histories, digital campaigns	Much richer affect and heritage evidence, but major inequalities in internet access, language resources, archiving and research capacity persist

The periodisation deliberately does not use conferences as automatic dividing lines. It treats 1988 as a conjunction of scientific, political, and media processes in particular settings; 1993 as the beginning of a repeatable survey infrastructure; 2007 as a media-digital and representational shift; and 2015 as a convergence of threshold politics, new affect measures, activism, and active heritage-making.

Anchor-source architecture and evidence hierarchy

An **anchor source** is not an "absolute source." It is a source sufficiently authoritative and transparent to support a specified claim within a defined scope.

An anchor should be scored against: authority; transparent methodology; stable identifier; version history; temporal continuity; geographic coverage; availability of original data or text; reproducibility; licensing; preservation stability; citation conventions; documented bias; and capacity to support the particular claim being made.

Evidence tiers

Tier	Definition	Permitted function
Tier 1: Primary authoritative anchor	Original observational dataset, official treaty, archived questionnaire and microdata, accession record, or primary historical document with stable provenance and methodology	May directly support a bounded factual claim
Tier 2: Reputable research dataset or scholarly reconstruction	Peer-reviewed dataset, harmonised series, or critical scholarly edition with transparent methods	May support analysis where limitations are stated and primary evidence is recoverable

Tier	Definition	Permitted function
Tier 3: Archival or media source	Newspaper, broadcast, film, catalogue, oral history, protest object, exhibition record, parliamentary speech	Supports representation, discourse, reception, participation, and heritage claims—not physical climate or population prevalence without another anchor
Tier 4: Discovery lead	Search index, snippet service, aggregator, secondary timeline, unsystematic bibliography	Locates candidates but should not carry a central claim alone
Tier 5: Unverified or unsuitable	Unprovenanced repost, undocumented chart, scraped content with unknown source population, altered image, unsupported statistic	Exclude or retain only in a quarantine list

Anchor source registry

All registry rows use the requested common schema. “Question wording” is not applicable to physical and institutional sources.

source_name	owner	source_type	coverage_start	coverage_end	geography	languages
HadCRUT5 [17]	UK Met Office Hadley Centre and UEA CRU	Tier 1 gridded observational reconstruction	1850	Present	Global land and ocean	English metadata
NASA GISTEMP v4 [18]	NASA GISS	Tier 1 global surface analysis	1880	Present	Global	English metadata

source_name	owner	source_type	coverage_start	coverage_end	geography	languages
ERA5 [19]	Copernicus Climate Change Service, implemented by ECMWF	Tier 1 reanalysis	1940	Present	Global	English interface and documentatio
NOAA GHCN-Daily [20]	NOAA National Centers for Environmental Information	Tier 1 station dataset	Station-dependent; nineteenth century for some stations	Present	Global station network	English metadata; station names worldwide
Berkeley Earth temperature data [21]	Berkeley Earth	Tier 2 research dataset	1750 land-only; 1850 land-ocean	Present	Global	English metadata
WMO State of the Global Climate and indicators [22]	World Meteorological Organization	Tier 1 institutional synthesis	Annual series; indicator-dependent	Present	Global	Multiple WMO UN languages varying by product
IPCC assessment archive [23]	IPCC/WMO/ UNEP	Tier 1 assessment-document corpus	1988 institution; reports from 1990	Present	Global assessment with uneven underlying literature coverage	UN languages for major summaries; report availability varies

source_name	owner	source_type	coverage_start	coverage_end	geography	languages
UNFCCC, Kyoto and Paris documentary corpus [24]	United Nations Framework Convention on Climate Change	Tier 1 legal and institutional corpus	1992	Present	Parties to UN climate regime	Six UN languages for core agreements
National meteorological series, selected	National meteorological agencies	Tier 1 national or regional observational record	Country-dependent	Present	National, territorial or station-level	National languages
Parliamentary and government report archives	Legislatures, ministries and national archives	Tier 1 or Tier 3 document corpus	Country-dependent	Present	National	National and parliamentary languages

Critical baseline rule. HadCRUT5 commonly expresses anomalies relative to 1961–1990, GISTEMP commonly uses 1951–1980, while WMO frequently communicates change relative to 1850–1900. These values cannot be concatenated, averaged, or visually compared as if zero meant the same physical temperature. Every derived series must be transformed to a documented common reference period using an overlap calculation or analysed as within-product deviations. [25]

A reproducible physical workflow should record the dataset name, exact version, retrieval date, original baseline, chosen common baseline, spatial mask, temporal aggregation, uncertainty field, transformation code, and checksum. Derived event metrics should separately record event definition—for example, percentile threshold, duration, station or grid choice—and should never infer emotional response merely from exposure.

Claim–source map

Claim	Required anchor	Supporting source	Sources that are insufficient alone
“Temperature rose during period X”	Versioned global or national physical dataset	WMO synthesis or peer-reviewed reconstruction	Newspaper description, survey belief, photograph of heat
“A location experienced an exceptional heat event”	GHCN-D, national station series or ERA5 with a stated event definition	Government event report, mortality or emergency records	Viral posts or an isolated anecdote

Claim	Required anchor	Supporting source	Sources that are insufficient alone
“Scientists could defend anthropogenic warming by date X”	Primary scientific text plus scholarly history	Later assessment tracing the state of knowledge	Modern popular timeline alone
“Government or international institutions recognised the issue”	Treaty, official report, hearing or decision	Scholarly institutional history	News report alone
“People in country X became more worried”	Repeated survey series with stable or linked wording, sampling and weights	Qualitative interviews or media reception research	Media volume, Google searches or policy support
“Climate was a more salient public issue”	Open-ended most-important-problem item or comparable agenda measure	News volume and search behaviour	Direct worry item alone
“Threat language increased in media”	Reproducibly sampled corpus, duplicate control and validated coding	GDELT, MeCCO or archive search counts	Hand-picked headlines
“An artefact acquired heritage value”	Accession, preservation, exhibition, reuse, community recognition or transmission evidence	Citation history, educational reuse, anniversary commemoration	Age, fame, digitisation, or online availability alone
“A global pattern exists”	Comparable multi-country evidence plus coverage and invariance audit	Parallel national series	Aggregated percentages from unrelated polls

Public-opinion evidence, direct affect, and index feasibility

The survey evidence is rich enough for comparative research but heterogeneous enough to make a single global fear series misleading. The central distinction is between **what respondents feel**, **what risks they perceive**, **how important they think the issue is**, and **what actions they support**.

Public-opinion dataset registry

| source_name | owner | source_type | coverage_start | coverage_end | geography | languages | unit_of_analysis | variables | question_wording | access | licence | version | stable_identifier | strengths | limitations | heritage_relevance | social_relevance | recommended_use | |---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---| | Early US climate polls, including 1981 and 1986 | Multiple polling organisations; identified through historical review | Tier 2 historical poll series/ reconstruction | 1981 | Late 1980s | United States | English | Respondent and poll | Awareness;

seriousness | 1986 poll asked whether global warming was a serious problem; exact original instrument must be recovered before coding | Often through reports, archives or secondary compilations | Poll-specific | Poll-specific | Poll report or archive record | Earliest located US public-recognition evidence | Sparse; inconsistent wording; incomplete microdata and technical documentation; seriousness is not fear | Medium as evidence of early public knowledge | High | Use to establish a pre-series chronology, not a continuous affect trend [26] | | Gallup Environment/global-warming worry series | Gallup | Tier 1/2 repeated national survey series | 1989 | Present | United States | English; Spanish administration may apply in recent surveys | Adult respondent | Personal worry; belief, timing, cause and policy items in some waves | "How much do you personally worry about global warming?" Responses include great deal, fair amount, only a little, not at all. | Published trends; detailed microdata access varies | Gallup terms; redistribution restrictions likely | Annual poll release/wave | Gallup question trend and poll dates | Longest clearly documented national direct-worry series; stable core wording | Modes, sampling frames and surrounding questionnaire change; only one country; "global warming" frame | High as a repeatedly reproduced public-opinion series | Very high | Primary US direct-affect anchor; retain all response categories and method changes [11] | | Eurobarometer environmental and climate surveys | European Commission; archived by GESIS | Tier 1 regional cross-national surveys | Relevant greenhouse items by 1989; dedicated recurring climate surveys from 2009 | Present | EU member states and, depending on wave, candidates or other European jurisdictions | National survey languages; core English/French documentation | Respondent | Awareness, seriousness, global priorities, responsibility, policy and behaviour | 1989 archived item: "Can you tell me if it is a very serious problem, quite serious or not very serious?" referring to the greenhouse effect/global warming | GESIS and EU survey portal; questionnaires and many datasets downloadable after registration | GESIS/EU data terms; attribution required | ZA study version and Eurobarometer wave | ZA study number and survey identifier | Large regional samples; national translations; repeated technical infrastructure | EU is not global; changing membership, wording, mode and translations; many items measure seriousness rather than emotion | High as European institutional-public memory | Very high | Regional comparison; build an item-level concordance rather than one uninterrupted score [27] | | ISSP Environment I–IV | International Social Survey Programme; GESIS archive | Tier 1 harmonised cross-national microdata | 1993 | 2020 | Participating countries, varying by wave | Numerous national languages; source questionnaire in English | Respondent | Environmental danger, knowledge, concern, willingness to pay, behaviour, trust and climate items | Earlier waves refer to greenhouse effect/global warming; 2010 and 2020 instructions specify "climate change" or nearest equivalent | Registration and download from GESIS; questionnaires and documentation available | Scientific-use and archive terms; country conditions may vary | 1993 v1.0.0; 2010 v3.0.0; 2020 v2.0 at research date | DOIs including 10.4232/1.2450, 10.4232/1.13271 and 10.4232/1.14153 | Best long-run harmonised environment series; microdata and national documentation | Country participation changes; probability design, mode and weights differ; core climate terminology changes; not primarily an affect scale | Very high | Very high | Main repeated cross-national risk/seriousness anchor, with wave-specific construct labels [28] | | GlobeScan international climate items | GlobeScan and commissioning partners | Tier 2 cross-national polling | At least 2000 for identified broad climate seriousness comparison | Later waves vary | 34 countries in cited 2000 study | National languages | Respondent | Seriousness of climate change/global warming | "How serious a problem do you consider climate change or global warming, due to the greenhouse effect?" | Reports readily available; microdata and questionnaires require investigation or permission | Proprietary unless deposited elsewhere | Survey-specific | Report and archive record | Broad country coverage for its period | Proprietary access; uncertain translation and microdata availability; seriousness is not fear | Medium | High | Historical comparator if original questionnaire, samples and permissions can be verified [29] | | Yale Climate Change in the American Mind | Yale Program on Climate Change Communication and George Mason University | Tier 2 repeated national research survey | 2008 | Present | United States; selected subnational estimates | English and, in some administrations, Spanish | Adult respondent | Belief, worry, perceived harm, risk, norms, efficacy, policy and audience segments | Core item: “How worried are you about global warming?” | Reports, visualisations, methods and selected data products | Project terms; seek permission for microdata redistribution | Survey wave | Report date, dataset or report DOI where assigned | Rich construct separation; roughly biannual national series; strong documentation | One country; web-panel and weighting methods require wave checks; not interchangeable with Gallup wording | High in communication research and education | Very high | US affect-mechanism analysis and validation against Gallup; do not pool automatically [30] | | Lloyd’s Register Foundation World Risk Poll | Lloyd’s Register Foundation and Gallup | Tier 2 near-global repeated survey | 2019 | 2023 data publicly identified; later cycles must be versioned separately | More than 140 countries in initial cycle | Many interview languages | Adult respondent | Threat perception, personal experience, safety and risk | “Do you think that climate change is a very serious threat, a somewhat serious threat, or not a threat at all…?” | Free downloadable data with attribution | Data-use and attribution terms | 2019, 2021 and 2023 releases | Release-level files and documentation | Exceptional geographic reach; repeated core risk framework; includes countries weakly represented elsewhere | Measures threat, not fear; telephone versus face-to-face designs and country coverage vary; short time series | Medium to high | Very high | Primary near-global threat-perception comparator, kept separate from worry/anxiety [31] | | Afrobarometer | Afrobarometer network and national partners | Tier 2 cross-national regional survey | Climate modules vary; broader series since 1999 | Present | Up to 42 African countries across rounds | English, French, Portuguese, Arabic and national/local interview languages | Adult respondent | Awareness, perceived causes, local impacts, responsibility, adaptation and action | Usually begins with whether the respondent has heard of climate change, followed by perceived effects; wording is round-specific | Public data after registration and embargo rules | Afrobarometer data-use terms | Round and country release | Round/country dataset identifiers | Essential African representation; face-to-face national probability surveys; local-language fieldwork | Climate questions are not present identically in every round; awareness filters alter denominators; little direct fear measurement | High potential for local memory and public history | Very high | Regional awareness and impact-perception analysis; never relabel filtered impact items as fear [32] | | Latinobarómetro | Corporación Latinobarómetro | Tier 2 regional survey | General series from 1995; climate items intermittent | Present | Approximately 18 Latin American countries | Spanish and Portuguese | Adult respondent | National problems, trust, environment and occasional climate items | Wave-specific; no continuous climate-fear item should be assumed | Questionnaires, technical sheets and some data files available under registration or terms | Organisation-specific | Annual study | Annual survey file and questionnaire | Long regional public-opinion infrastructure; useful open-ended salience measures | Climate content is intermittent; national samples and methods change; not a direct climate-worry series | Medium to high | Very high | Discovery and salience source; include only verified climate variables [33] | | Asian Barometer Survey | Asian Barometer Survey network | Tier 2 regional cross-national survey | Climate-specific additions particularly visible in Wave 6 | Present | East, Southeast and parts of South Asia, varying by wave | National languages | Adult respondent | Governance, values, environment and newer climate questions | Wave-specific; exact wording must be extracted from country questionnaires | Application or project download conditions | Project-specific | Wave and country | Dataset and questionnaire identifiers | Important non-Western political and cultural comparison | Climate series is recent and not longitudinally continuous; participating countries vary | Medium | High | Supplementary Asia comparison, not a long-run anchor [34] | | Australian Climate of the Nation | The Australia Institute | Tier 2 national report series | 2007 | Present | Australia | English | Adult respondent | Concern, perceived impacts, policy, energy and responsibility | Wave-specific | Public reports; underlying microdata not routinely open | Copyrighted reports; permission needed for reproduction or data reuse | Annual report | Report edition | Long-running Australia-focused series | Proprietary microdata; questionnaire and mode changes require direct documentation | High national reuse | Very high | National trend context; use only documented repeated items [35] | | Lowy Institute Poll climate item | Lowy Institute | Tier 2 national repeated poll | Relevant climate item repeated across multiple years | Present | Australia | English | Adult respondent | Urgency, costs and policy preference | Respondents choose among positions including that warming is a serious and pressing problem requiring action despite significant costs | Reports and interactive trend pages; microdata access varies | Lowy terms | Annual poll | Poll report and year | Stable and influential Australian policy trade-off item | Mixes seriousness with cost-bearing policy preference; cannot be interpreted as fear | High public and political reuse | Very high | Policy-preference indicator, explicitly excluded from direct-affect series [36] | | Climate Change Anxiety Scale | Original academic authors; subsequent adapters | Tier 2 validated psychometric instrument | 2020 | Present | Initially US; subsequently translated and tested in multiple countries | Multiple validated translations | Individual scale response | Cognitive-emotional impairment and functional impairment | Thirteen core items concerning intrusive thoughts, concentration, sleep, crying and effects on functioning | Journal appendices and replication studies; instrument-use conditions should be checked | Publication/instrument-specific | Original and adapted versions | Article DOI and scale citation | Separates distress and impairment; widely used; psychometric literature growing | Not a historical public-opinion measure; convenience samples common; adaptation does not guarantee scalar invariance | Medium as evidence of the recent naming of climate anxiety | High in mental-health and youth discourse | Analyse as a recent validated scale, separate from ordinary worry [37] | | Climate Change Worry Scale | Academic authors | Tier 2 validated psychometric instrument | 2021 | Present | Initially study-specific; adaptations emerging | English and translated versions | Individual scale response | Persistent climate worry | Ten-item self-report worry measure | Article and supplementary materials | Publication-specific | Original/adapted | Article DOI | More directly targeted at worry than broad anxiety impairment | Short historical span; sample and cultural validation limitations | Medium | High | Contemporary affect module; do not back-project into earlier surveys [38] |

Direct evidence versus proxy matrix

Measurement class	Typical wording or variable	What it validly supports	Can it be called "fear"?
Direct affect evidence	Afraid, fearful, frightened, worried, anxious, distressed, dread	Prevalence or intensity of the named feeling, subject to translation and scale validity	**Yes, but use the exact affect term.** Worry is not automatically identical to fear.
Risk perception	Likelihood of harm, perceived threat, vulnerability, expected personal or national effects	Cognitive appraisal of danger or exposure	**No.** It may correlate with fear but is not the same construct.
Concern or seriousness	Very serious problem, concerned, importance of environmental danger	Evaluative concern or problem severity	**Usually no.** "Concern" may contain affective content but should remain its own category.
Issue salience	Most important problem, priority ranking, spontaneous mention	Position on the public agenda	**No.** Salience can rise without stronger personal fear.

Measurement class	Typical wording or variable	What it validly supports	Can it be called "fear"?
Policy preference	Support emissions cuts, pay more, act now despite costs	Political preference and willingness to accept policy consequences	**No.** Support may arise from values, duty, efficacy, ideology, or interest.
Behavioural intention or behaviour	Reduce travel, vote, protest, purchase, conserve energy	Intended or reported action	**No.** Behaviour is not an emotion and may be driven by many motives.
Functional impairment	Sleep disruption, concentration difficulty, interference with work or relationships	Severity and consequences of climate-related distress	**Not by itself.** It should be reported as impairment or clinically relevant distress, not general population fear.
Media proxy	Frequency of "fear," "crisis," "catastrophe," negative tone	Representational prevalence in a specified corpus	**No population inference.** It measures media language, not audience emotion.
Search or social-media proxy	Query volume, hashtags, engagement, sentiment model	Attention or platform-specific expression	**No.** Search behaviour and automated sentiment are neither probability samples nor stable affect measures.
Physical exposure	Temperature anomaly, heatwave days, mortality or disaster loss	Environmental or health context	**No.** Exposure must be linked to independent affect evidence.

Global Fear Index decision: Option 3—build multiple parallel indicators rather than one index.

A single index would require evidence that the contributing questions measure the same latent construct across languages, countries, modes, and decades. That condition is not met. "How worried are you?", "How serious is the problem?", "Is it a threat?", "Do you support action?", and "Does it impair your functioning?" are different constructs. Converting their response scales to 0–100 would create numerical similarity without conceptual equivalence.

Cross-national climate-anxiety research has found support for some levels of invariance in selected samples—for example configural and metric invariance across China, India, Japan and the United States—but this does not establish universal scalar invariance or justify combining all existing poll items into one world mean. [39]

The recommended indicator family is:

Indicator	Included evidence	Excluded evidence
Direct climate affect	Fear, worry, anxiety items with explicit climate referent	Seriousness, belief, policy and media tone

Indicator	Included evidence	Excluded evidence
Climate threat appraisal	Personal, national or global threat and likelihood items	Affect and policy support
Climate seriousness/ concern	Seriousness and concern scales	Fear unless named
Climate issue salience	Open-ended priority and ranking measures	Prompted worry
Climate-action preference	Mitigation, adaptation, willingness-to-pay and cost trade-offs	Affect
Climate-distress impairment	Validated impairment scales	Ordinary concern and threat
Representational intensity	Media crisis, catastrophe and affect vocabulary	Claims about population emotion

Within a survey family and a demonstrably invariant subset of countries, a regional or wave-limited latent index may be tested. Missing years should not be filled by media coverage or linear interpolation across long gaps. Gaps should remain explicit, or be modelled only in a clearly labelled statistical analysis with sensitivity intervals.

Media, language, cultural records, and heritage evidence

Media archives and computational datasets differ fundamentally in what they preserve. Some store original full text or video; some expose captions; some provide only metadata, counts, n-grams, themes or snippets; and some are discovery tools that lead to copyrighted material elsewhere.

Media and cultural source registry

| source_name | owner | source_type | coverage_start | coverage_end | geography | languages | unit_of_analysis | variables | question_wording | access | licence | version | stable_identifier | strengths | limitations | heritage_relevance | social_relevance | recommended_use | |---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---| | GDELT Project [40] | GDELT Project | Tier 2 computational dataset and Tier 4 discovery system | Event database from 1979; GDELT 2.0 architecture from 2015 | Present | Global claim; actual source density varies | Monitors more than 100 languages, often using machine translation | Article mention, event, actor, theme, location, tone and time interval | Events, themes, entities, geolocation, tone and source metadata | N/A | Bulk files and Google BigQuery | Open analytical access; original article copyrights remain with publishers | GDELT 1.0, 2.0 and product-specific releases | Dataset table and file identifiers | Scale, multilingual reach, rapid updates, reproducible querying | Source list and ingestion change; duplicate/syndicated stories; machine translation; full original text often unavailable; 1979 continuity differs by product | Medium as a map of news traces | High | Discovery, aggregate attention and candidate sampling; never treat tone as public fear | | Internet Archive TV News Archive [41] | Internet Archive and contributing broadcasters/archives | Tier 3 audiovisual archive and caption search | Collection-dependent; searchable modern US coverage especially dense | Present | Primarily US broadcast holdings with selected international material | Predominantly English; other languages uneven | Broadcast, programme, clip and caption segment | Captions, programme metadata, time and channel | N/A | Caption search, viewing and clip tools | Research access does not erase broadcaster copyright; public display and redistribution require review | Continuously growing | Archive item identifier | Searchable

broadcast evidence; preserves audiovisual framing and repetition | Closed-caption errors; network and period bias; clip availability and reuse restrictions | High | Very high | Selected broadcast case corpus; preserve programme, timestamp and item ID | | Media and Climate Change Observatory | University of Colorado-led MeCCO collaboration | Tier 2 reproducible monitoring dataset | Many series from 2000 or 2004 | Present | 59 countries and seven regions in current monitoring | 14 languages in current description | Outlet-month and article count; selected broadcast units | Volume of “climate change” or “global warming” coverage | N/A | Downloadable datasets and monthly summaries; underlying commercial databases may require subscriptions | Dataset-specific; source article text governed by Factiva, Nexis and other vendors | Series-specific | Dataset DOIs, including country and regional series | Transparent, long-running climate-specific monitoring; stable published datasets | Selected outlets only; commercial database dependence; counts do not measure content or public emotion | High as a longitudinal media-monitoring record | Very high | Primary multinational media-volume anchor from the 2000s onward [42] | | National newspaper archives: Trove, Chronicling America, national libraries and licensed databases | National libraries, archives and commercial providers | Tier 3 primary-source repositories | Country and title dependent, often nineteenth century onward | Varies | National or regional | National publication languages | Article, page, issue and title | Full text/OCR, image, date, title and metadata | N/A | Open, reading-room or subscription access | Public-domain, library, publisher and database terms vary | Platform and collection release | Newspaper title, issue and page identifier | Long historical depth; original layout and advertisements; national specificity | Unequal digitisation; OCR errors; missing titles; colonial collection structures; paywalls; changed newspaper markets | Very high | Very high | Deep-case primary corpus after title-level coverage and rights audit | | Parliamentary and broadcast archives | Legislatures, public broadcasters and national archives | Tier 3 primary repositories | Country-dependent | Present | National | National languages | Speech, debate, question, hearing, programme or segment | Text, video, speaker, party, date and topic | N/A | Official search portals, catalogues or reading rooms | Jurisdiction-specific | Session or programme record | Official document or audiovisual ID | Direct institutional and public-communication evidence | Search and transcription quality vary; state-media contexts may constrain viewpoint diversity | High | High | Link institutional milestones to media circulation and later reuse | | Google Books Ngram Viewer and downloadable n-grams | Google Books; downloadable corpus mirrors | Tier 4 aggregate discovery tool | Common research corpus begins around 1800 | Corpus edition endpoint | Selected book corpora | Major corpus languages, not globally balanced | N-gram-year frequency | Token or phrase frequency normalised by corpus size | N/A | Viewer and downloadable n-gram files | Dataset terms; underlying books retain copyright | Corpus edition | Corpus and n-gram file identifier | Fast long-run terminology exploration | No periodicals; corpus composition changes; OCR and metadata errors; cannot inspect many underlying passages; strong limits on cultural inference | Medium | Medium | Locate candidate terminology shifts only; confirm with full-text sources [43] | | Web archives, including national web archives and the Wayback Machine | Internet Archive and national collecting institutions | Tier 3 repository and Tier 4 discovery system | Mid-1990s onward, uneven | Present | Global but highly uneven | Web languages | Archived URL snapshot | HTML, images, linked files, date and capture metadata | N/A | Public interface and, in some contexts, research APIs or reading rooms | Original copyright persists; robots exclusions and access restrictions apply | Capture-specific | Archived URL and timestamp | Essential for vanished campaigns, government pages and digital exhibitions | Selective capture, missing assets, dynamic-page failure, link rot and jurisdictional restrictions | Very high for born-digital heritage | Very high | Preserve campaign and institutional web histories; cite capture timestamp | | Social-media research datasets | Platform, researchers or archives | Tier 2–4, depending on method and preservation | Platform-dependent | Platform-dependent | Global claim but platform penetration unequal | Multilingual, with major NLP-resource imbalance | Post, account, hashtag, image or interaction | Text, engagement, network, time and inferred attributes | N/A | Often restricted APIs, hydrated IDs or controlled access | Platform terms, privacy law and research-consent constraints | Dataset/API version | DOI where deposited, otherwise post ID | Captures vernacular expression and mobilisation | Deleted content; demographic non-representativeness; API instability; automated sentiment bias; ethical and consent concerns | Potentially high but fragile | Very high | Small, ethically reviewed born-digital sample only; no population fear estimate | | Documentary-film and cultural-production databases | Film archives, broadcasters, distributors and specialist databases | Tier 3 index or repository | Film-dependent | Present | International, with distribution bias | Multiple | Film, episode, screening, review and promotional object | Title, synopsis, release, audience and production metadata | N/A | Catalogue, archive, streaming or licensed copy | Copyrighted; quotation and clip display restricted | Edition or release | Catalogue ID, ISBN/ISAN where available | Connects climate narratives with popular culture and later educational reuse | Databases often index rather than preserve; distribution and audience data incomplete; research favours high-profile films | High | High | Curated artefact corpus, not a prevalence estimate [44] | | Climate advertising and visual-communication collections | Museums, NGOs, advertising archives, libraries and researchers | Tier 3 collection | Collection-dependent | Present | Uneven | Multiple | Advertisement, poster, photograph, infographic or campaign | Image, slogan, sponsor, audience, medium, date and rights | N/A | Catalogue or permission-based access | High rights complexity; moral rights, trademarks and donor restrictions | Collection-specific | Accession or catalogue ID | Strong evidence of visual affect and persuasion | Provenance and circulation often unclear; repeated online images may be decontextualised | Very high | High | Select only items with provenance, circulation context and display permission |

Syndication and duplication rule. News records must be clustered by near-identical text, wire-service attribution, headline similarity, publication time, and embedded media. A hundred syndicated copies are evidence of distribution reach, not a hundred independently produced interpretations.

Language rule. Searches should use historically appropriate term families rather than direct translations of one modern English lexicon. For example, a corpus may need separate searches for equivalents of greenhouse effect, climatic change, global warming, climate crisis, emergency, dangerous heat, abnormal heat, drought, scorching weather, ecological breakdown, anxiety, worry and threat. Historical dictionaries, native-speaker review, back-translation, and test retrieval should precede quantitative comparison.

Heritage-evidence registry

| source_name | owner | source_type | coverage_start | coverage_end | geography | languages | unit_of_analysis | variables | question_wording | access | licence | version | stable_identifier | strengths | limitations | heritage_relevance | social_relevance | recommended_use | |---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---| | Charles David Keeling Papers | UC San Diego Library/Scripps archives | Tier 1 archival finding aid and collection | 1934 | 2006, bulk 1955–2000 | United States with global scientific networks | Primarily English | Folder, correspondence, report, data documentation and lecture material | Scientific work, institutional correspondence and Scripps CO_2 programme records | N/A | Finding aid online; materials subject to archive access rules | Archive and donor restrictions; reproduction permission required | Collection arrangement | ARK finding-aid identifier | Formal preservation of foundational monitoring practice and institutional history | Does not alone establish mass public memory; collection description may not represent every item | Very high institutional preservation and later scholarly reuse | Medium to high | Anchor scientific-heritage dossier [45] | | WMO International Meteorological Organization Legacy Collection | WMO Archives; UNESCO Memory of the World | Tier 1 formally recognised documentary heritage | Nineteenth-century international meteorological cooperation | Historical collection; inscribed 2025 | International | Multiple working languages in records | Archival document and collection | Data exchange, organisational history and meteorological cooperation | N/A | WMO archive/catalogue conditions | WMO/UNESCO and item-specific conditions | 2025 inscription record | UNESCO/WMO register record | Formal international recognition; demonstrates weather-data cooperation as world documentary

heritage | It concerns meteorological institutional history more broadly, not public fear specifically | Very high formal recognition and preservation | High | Heritage-persistence milestone for the instrumental and institutional clock [46] | | Bristol Museums climate-movement collecting | Bristol Culture/Bristol Museums | Tier 1 museum collecting record | Artefacts primarily from 2018–2019 movements | Continuing | Bristol and wider UK movement links | English | Accessioned object, photograph, placard and collecting narrative | Provenance, movement affiliation and museum rationale | N/A | Museum catalogue/blog and permission-based object access | Museum and creator rights | Accession-specific | Museum accession or collection record | Explicit contemporary collecting rationale; documents youth strikes and Extinction Rebellion | Local and institution-selected; collecting decision may privilege highly visible movements | Very high institutional preservation and later display | Very high participation and community recognition | Model for how to document movement artefacts as heritage [47] | | Mallee Climate Oral History Collection | Museums Victoria and project partners | Tier 1 oral-history and museum collection | Memories cover long drought histories; project-era recording | Collection date onward | Mallee region, Australia | English; community language context must be checked | Interview, photograph and participant record | Lived drought, adaptation, endurance and perceptions of climate change | Interview guide and consent are project-specific | Museum access and oral-history conditions | Consent, copyright and cultural restrictions apply | Collection-specific | Museum collection identifier | Connects climate perception, lived experience, place and preservation | Not a probability sample; narration shaped by interview context; sensitive local knowledge | Very high community-memory evidence | Very high | Core Australian heritage case, contingent on consent and reuse permissions [48] | | NOAA Voices Oral History Archives | NOAA and partner projects | Tier 2/3 aggregated oral-history archive | Collection-dependent | Present | United States, coasts, oceans and some Indigenous/local communities | Primarily English; collection-dependent | Interview and collection | Environmental change, climate, fisheries, coasts and disasters | Interview schedules vary | Searchable digital archive | Interview-specific consent and reuse rules | Continuously expanding | Interview and collection identifiers | Approximately 2,300 interviews in 118 collections; strong human-experience and education potential | Aggregates heterogeneous projects; community rights and interview consent must be checked individually | Very high | Very high | Comparative oral-history discovery source, not uniform data [49] | | Extinction Rebellion York Archive | University of York/Archives Hub description | Tier 1 archival collection record | Movement material from late 2010s | Collection-dependent | York, UK | English | Document, bill, leaflet and organisational record | Emergency declaration demands and movement activity | N/A | Catalogue and archive access | Archive and creator restrictions | Fonds-specific | GB 193 XREB identifier | Demonstrates transition from live activism to formal archival custody | Local branch; collection completeness and donor selection require examination | Very high | High | Movement-heritage case and comparison with museum collecting [50] | | UNESCO Memory of the World Register | UNESCO | Tier 1 heritage register | Programme from 1990s; entries cover much earlier material | Present | International, but nominations and inscription capacity uneven | UNESCO languages | Registered documentary collection | Significance, provenance, preservation and access | N/A | Public register | UNESCO and holding-institution conditions | Register year | Entry identifier | Formalised evidence of documentary-heritage recognition | Nomination-based; underrepresentation does not imply lack of heritage value; inscription is not global social consensus | Very high formal recognition | Medium to high | Formal recognition variable, never the sole heritage criterion [51] | | Local and Indigenous environmental-knowledge collections | Community organisations, cultural centres, Indigenous nations and partner repositories | Tier depends on community authority and provenance | Knowledge may be intergenerational; documentation date varies | Continuing | Community-defined | Community languages | Narrative, practice, seasonal knowledge, place, image or restricted record | Environmental indicators, adaptation, obligations and memory | Community-defined protocols, not standard survey wording | Community-controlled or restricted | Community authority overrides assumed openness | Collection-specific | Community or repository identifier; Local Contexts labels where applied | Essential for non-state, intergenerational environmental knowledge and stewardship | High risk of extraction, decontextualisation and inappropriate disclosure; some knowledge should not be digitised or publicly indexed | Potentially very high, under community definition | Very high | Include only through partnership, authority to control and negotiated access; never treat absence from public catalogues as absence of heritage [52] |

Heritage evidence should be coded along six independent dimensions:

Dimension	Strong evidence	Weak or insufficient evidence
Social reach	Documented audience, circulation, screenings, classroom adoption	Assumed importance based on online presence
Social participation	Community creation, annotation, donation, protest use or oral-history involvement	Institution collecting an object without creator engagement
Community recognition	Community-defined significance, custodianship, ceremony or protocol	External researcher assigning significance
Institutional preservation	Accession, catalogue record, conservation plan, persistent identifier	Temporary exhibition or unstable webpage
Intergenerational transmission	Repeated teaching, family/community narration, curriculum or practice	One-time publication
Later reuse	Anniversary coverage, remounting, reproduction, quotation, litigation, activism or education	Passive storage with no evidence of reuse
Formal recognition	Heritage register, official designation or documented collection policy	Fame or age alone

A heritage claim should normally require evidence in at least **two dimensions**, one of which should be preservation, community recognition, transmission, or later reuse. Formal designation is not mandatory: community-controlled heritage may be highly significant without state or UNESCO recognition.

Global coverage, bias audit, and recommended scope

Global coverage and bias map

Bias dimension	Likely distortion	Required mitigation
English-language dominance	Makes Anglophone terms, media breakthroughs and academic interpretations appear earlier or more universal	Separate language chronologies; employ native-speaker validation; report untranslated and inaccessible corpora

Bias dimension	Likely distortion	Required mitigation
Western survey dominance	Overstates countries with long commercial and academic polling infrastructures	Distinguish absence of evidence from evidence of absence; add Afrobarometer, regional barometers and World Risk Poll without forcing construct equivalence
Unequal archive digitisation	Countries with searchable newspapers appear more publicly engaged	Record title-years available, missing issues, OCR quality and analogue-only holdings
Colonial archive structure	Centres administrative observers and marginalises colonised or Indigenous knowledge holders	Trace provenance and collecting power; seek community repositories and counter-archives
Weak polling infrastructure	Produces irregular samples, urban bias or missing conflict-affected populations	Retain design variables; use survey-quality flags; avoid national rankings where coverage is weak
Translation inequivalence	“Fear,” “worry,” “concern,” “threat” and “seriousness” may not map one-to-one	Store original wording, literal and idiomatic translation, translator notes, response scale and cognitive-test evidence
Cultural meaning of emotion	Public expression norms alter willingness to select strong affect labels	Test differential item functioning and measurement invariance; supplement surveys with qualitative interpretation
State media and censorship	Low visible contestation may reflect control rather than low concern	Include media-system metadata and non-news cultural or diasporic sources
Uneven physical exposure	National means obscure local heat, drought, humidity, urban heat and livelihood vulnerability	Link respondents or artefacts only at an ethically and statistically appropriate spatial scale
Internet-access inequality	Digital traces overrepresent connected, young, urban and platform-active groups	Treat social media and searches as platform evidence, not population measures
Small-island-state underrepresentation	Global averages can erase acute existential and cultural concerns	Include a SIDS sentinel dossier even if it cannot support a full survey time series
Indigenous knowledge exclusion	Public archives may omit or restrict community knowledge for legitimate reasons	Apply CARE and community authority; record “restricted by protocol” rather than “missing” [53]
Destroyed or inaccessible archives	Creates false historical silence	Maintain a structured missingness register including destruction, closure, paywall, censorship and unknown survival

Scope-model comparison

Model	Feasibility	Originality	Representativeness	Access and language demand	Computational demand	Ethical risk	He ar
Global Metadata Backbone + Three Regional Deep Cases	**High**, if the global layer stores metadata rather than full content	High: combines multiple clocks and heritage evidence	Moderate; representativeness is transparent rather than claimed	Manageable with English plus one additional research language and selective translation support	Moderate	Moderate; can isolate sensitive community collections	**St** be de pe pr re cu pa
Five to Six Stratified Cross-Regional Cases	Low to moderate for a Master thesis	Very high	Potentially highest	High: archives, polling systems, languages and ethics multiply rapidly	High	High, particularly if Indigenous or community data are treated superficially	Mo br ma dis clo pr wo he cla re
Multilingual Comparative Corridor	Moderate	Very high for language and conceptual history	Low to moderate globally, strong for translation comparison	High linguistic demand but bounded geography	Moderate	Moderate	Str co fol cir tra an ins ex

Recommended model: Global Metadata Backbone + Three Regional Deep Cases.

The most defensible case configuration is a **stratified data-ecology design**, not a claim that three countries represent humanity:

Deep case	Why it belongs	Principal sources	Principal limitation
United States, English	Long direct-worry series, 1988 public breakthrough, rich broadcast, congressional and archival records	Gallup, Yale, Hansen hearing, GISTEMP, Internet Archive TV News, national press	Exceptionally dense sources may dominate the conceptual model

Deep case	Why it belongs	Principal sources	Principal limitation
Germany or a German-centred European case, German plus English documentation	Different political-media trajectory; Eurobarometer and ISSP participation; strong institutional and newspaper infrastructure	ISSP, Eurobarometer, MeCCO German series, parliamentary and newspaper archives	Requires German-language competence and careful East/West historical treatment
Australia with a bounded community-heritage component, English plus community-defined language protocols	Strong meteorological history, recurring national polls, severe heat context, Trove and museum/oral-history evidence	Bureau of Meteorology/CSIRO, Lowy, Climate of the Nation, Trove, Mallee collection	Still Anglophone and settler-colonial; Indigenous evidence cannot be added as a token "perspective"

To reduce Western concentration, the global backbone should include two **sentinel gap dossiers**, not full fourth and fifth cases: one for India as a large multilingual media and survey environment, and one for a Pacific small-island state such as Fiji. Their purpose would be to test whether the metadata schema and inclusion rules fail when polling is sparse, archives are incomplete, climate impacts are acute, or heritage is community-controlled.

The alternative **multilingual corridor** would be preferable if the student has strong Spanish or another relevant language. A viable corridor could follow English–German–Spanish terminology and institutions across the United States or Australia, Germany, Spain, and one Latin American collection. It would be more original in translation history but less globally representative.

Minimum viable evidence base and pilot corpus

The Minimum Viable Evidence Base should be large enough to test every major claim type but small enough to permit provenance checking, wording harmonisation, rights review, and close historical interpretation.

Recommended evidence base

Layer	Minimum evidence	Recommended quantity	Sampling rationale	Access and likely cleaning cost
Physical climate	HadCRUT5 plus ERA5; one official national series per deep case; GHCN-D only for selected events	Two global products, three national products, six to twelve event extracts	One long global trend anchor, one event-resolution product, and local validation are sufficient; more global datasets add robustness but not new claim types	Low for annual global data; medium for ERA5; medium-high for station metadata and event definitions

Layer	Minimum evidence	Recommended quantity	Sampling rationale	Access and likely cleaning cost
International institutions	IPCC founding and selected assessment texts; UNFCCC, Kyoto and Paris; selected threshold documents	Approximately 25–35 documents	Sample only documents that change attribution, risk language, thresholds, institutional authority or later reuse	Low to medium; major work is document segmentation, versioning and multilingual alignment
National political institutions	Key hearings, government reports and speeches in each case	Approximately 8–12 per case	Select around turning points identified independently by scientific, media and survey evidence	Medium; parliamentary OCR and speech-version reconciliation may be substantial
Public opinion	Gallup/Yale for US; ISSP and Eurobarometer for cross-national comparison; one Australian series; World Risk Poll for near-global threat	Four to six survey families, eight to twelve focal waves, eight to fifteen harmonised variables	The aim is construct contrast and selected longitudinal comparison, not accumulation of all poll questions	Medium-high; exact wording, translations, weights, mode and missing-value harmonisation are the costliest tasks
Media	Two outlets or broadcast sources per case across four periods; aggregate MeCCO or GDELT context	**360–480 manually or semi-automatically coded items**	Three cases × four periods × two outlets × 15–20 items. This supports period and case comparison without making manual coding unmanageable	High; OCR, duplicate control, archive access, translation and copyright review dominate
Cultural artefacts	Films, advertisements, posters, educational materials, iconic graphics or photographs	30–45 artefacts	Ten to fifteen per case, purposively stratified by medium and period	Medium-high; provenance and image-display rights are often harder than discovery

Layer	Minimum evidence	Recommended quantity	Sampling rationale	Access and likely cleaning cost
Heritage-making evidence	Accession records, exhibitions, oral histories, curricula, anniversary reuse and community projects	15–25 dossiers	Five to eight per case, with at least two independent heritage dimensions per dossier	High interpretive cost; legal and ethical review may exceed coding time
Missingness and bias registry	Archive gaps, unavailable microdata, language gaps, destroyed collections, paywalls, protocol restrictions	One structured record for every source family and case-period cell	Missingness is itself a result and prevents false globalisation	Moderate; requires disciplined documentation rather than large data volume

Proposed pilot corpus

The first pilot should test the architecture before full collection:

Pilot component	Proposed sample
Temperature	HadCRUT5 annual and monthly series, rebased to 1850–1900 for communication; ERA5 extracts for one major heat episode per case; one national station or gridded series per case
Scientific history	Fourier, Arrhenius, Callendar, Charney-era assessment, IPCC establishment, First Assessment Report, one attribution passage from each subsequent assessment cycle
Institutional history	Hansen hearing, UNFCCC adoption, Kyoto, Paris, one 2°C document, one 1.5°C document and one climate-emergency declaration per case
Surveys	Gallup 1989 and selected later repeated waves; Yale selected worry waves; ISSP 1993, 2000, 2010 and 2020; one Eurobarometer early item and one dedicated climate wave; World Risk Poll 2019 and 2023; one Australian repeated item
Media	**120 items initially:** three cases × four historical periods × ten items, balanced across at least two outlets or media types
Cultural artefacts	Twelve pilot objects: four per case, covering at least three media such as newspaper front page, broadcast, film, poster, advertisement or curriculum
Heritage evidence	Nine dossiers: one scientific archive, one movement collection and one oral-history, educational, or community-memory object per case
Gap audit	India and Fiji metadata probes covering survey availability, newspaper access, language, heritage catalogues, and community-control issues

The four pilot periods should be:

1. 1988–1992, for public and institutional breakthroughs;
2. 1997–2001, for Kyoto and comparative-survey consolidation;
3. 2007–2010, for digital, documentary and assessment visibility;
4. 2018–2021, for 1.5°C, emergency language, youth mobilisation and climate-anxiety measurement.

The media sample should not simply take the first search results. For each outlet-period stratum, create a complete result list within fixed search dates, remove duplicates and syndicated copies, then draw a reproducible random or systematic sample. Purposively famous objects may be added as a separate "canonical artefact" stratum and must not be included when estimating ordinary content frequencies.

Recommended database structure

Entity	Required core fields
Source	Creator/owner, title, date, source type, language, geography, identifier, version, access date, licence and preservation location
Claim	Exact proposition, claim class, spatial and temporal scope, evidentiary threshold and confidence
Survey item	Original wording, translated wording, affect class, response scale, universe, mode, sample design, fieldwork, weighting and variable history
Climate observation	Dataset, variable, resolution, spatial unit, baseline, version, uncertainty and transformation
Media record	Outlet, publication/broadcast date, medium, search frame, full-text status, syndication cluster and coding fields
Heritage event	Accession, exhibition, reuse, curriculum inclusion, anniversary, community recognition, transmission or formal designation
Object–event link	Relationship type, date, evidence source and certainty
Rights and ethics	Copyright, access restriction, consent, cultural protocol, display permission, takedown procedure and community authority
Missingness	Not collected, inaccessible, destroyed, un-digitised, censored, paywalled, language-unprocessed, restricted by protocol or unknown

The database should preserve **original text separately from analytic classification**. It should never overwrite "worried" with a harmonised label such as "fear." Instead, it should store the source term, translated term, semantic class, and analyst's comparability decision.

Data-access and licensing risks

The principal risk is not inability to discover material; it is inability to reproduce or publicly display it. Climate datasets are generally favourable for research use, although attribution and versioning differ. ERA5 products are now generally available under CC BY terms in the Climate Data Store, while

HadCRUT5 uses the UK Open Government Licence and NASA government data are generally public-domain, subject to media-specific credits. [54]

Survey reports are often public while microdata, variable labels, and commercial trend tables are not. GESIS-hosted ISSP and Eurobarometer data are comparatively strong, but users must retain archive versions and terms. Gallup, proprietary Australian polls, and GlobeScan may permit quotation of published results without permitting redistribution of respondent-level data.

Newspaper databases frequently permit reading and text analysis but prohibit bulk redistribution of article text. A public archive should therefore publish metadata, derived features, short lawful excerpts, and links or identifiers—not a republished corpus—unless explicit permission or public-domain status is established.

Oral histories and community collections require item-level consent review. “Publicly searchable” does not necessarily mean consent for automated text mining, republication, geolocation, or decontextualised quotation. Restricted microdata should remain in approved environments; ICPSR, for example, distinguishes public-use files from restricted data managed according to consent and disclosure risk. [55]

Indigenous and community-controlled knowledge must be governed by **Collective Benefit, Authority to Control, Responsibility and Ethics**, not by an assumption that FAIR or open-data principles always take precedence. Local Contexts Traditional Knowledge Labels can communicate community-specific access, attribution, seasonal, sacred, gendered or reuse protocols. [52]

The project should include a takedown and correction procedure, permit “metadata only” and “not publicly displayable” records, and avoid publishing precise locations or culturally restricted content. Community records should be described in terms agreed with custodians, and the database should support community-preferred names and identifiers.

Revised questions and database-construction decision

Revised main research question

> **How did rising temperature and projected climatic warming become measurable, affectively meaningful, publicly contestable, and heritage-worthy across selected national and language contexts from 1988 to 2026, and what evidentiary limits govern the construction of a globally extensible digital cultural-heritage archive?**

This question is preferable to “When did the world become afraid of global warming?” because it does not presume a universal emotional subject, a single beginning, or equivalence among scientific knowledge, media attention, political action, and fear.

Recommended sub-questions

Sub-question	Principal evidence
How did changing observational systems and anomaly conventions shape what could be said about rising temperature at global, national and event scales?	HadCRUT5, GISTEMP, ERA5, GHCN-D and national meteorological datasets

Sub-question	Principal evidence
When and through which scientific and institutional processes did anthropogenic warming, numerical thresholds and emergency language become authoritative?	Scientific papers, IPCC reports, treaties, government reports, hearings and speeches
How have fear, worry, anxiety, seriousness, threat, salience and policy preference been operationalised differently across surveys, languages and periods?	Questionnaires, microdata, translations, sampling and psychometric studies
How did selected media systems and cultural forms represent warming, heat and climatic threat, and how did those representations differ across the case studies?	Reproducibly sampled newspapers, broadcasts, films, advertisements and educational materials
Which scientific records, media objects, protest materials and community memories have been preserved, reused or transmitted as climate heritage, and by whom?	Accession files, exhibitions, oral histories, curricula, community projects and reuse records
What geographic, linguistic, archival and ethical absences prevent the archive from being described as globally representative?	Coverage matrix, missingness register, rights review and sentinel-case audit

GO / MODIFY / NO-GO decision

Component	Decision	Reason
Global physical-temperature backbone	**GO**	Multiple authoritative, versioned datasets support global, regional and event-level context
Scientific and institutional chronology	**GO**	Dense primary documentation and strong scholarly interpretation exist from the nineteenth century onward
Cross-national public-opinion comparison	**MODIFY**	Proceed item-by-item and construct-by-construct; do not assume fear equivalence
Global media corpus	**MODIFY**	Build selected outlet-country-language strata plus metadata aggregates, not an exhaustive world corpus
Global heritage archive	**MODIFY**	Construct an extensible registry of documented preservation and reuse; do not equate institutional accession with all heritage
Indigenous and community knowledge layer	**MODIFY / partnership-dependent**	Inclusion requires community authority, negotiated purpose and culturally appropriate access
Single Global Fear Index	**NO-GO**	Construct, wording, scale, language, mode and invariance requirements are not met

Component	Decision	Reason
Parallel affect, threat, seriousness, salience, policy and impairment indicators	**GO**	Preserves conceptual validity and permits bounded comparisons
Full public redistribution of source texts and images	**NO-GO by default**	Copyright, database rights, consent and cultural protocols require source-specific permissions
Metadata-first linked database pilot	**GO**	Technically and academically feasible within a Master thesis if limited to three deep cases and a 120-item initial media pilot
Claim of a globally representative archive	**NO-GO**	Unequal archives, surveys, languages, internet access and community visibility preclude such a claim
Claim of a globally extensible archive with explicit gap documentation	**GO**	The schema can accommodate new regions and sources while recording uncertainty and structural absence

The final decision is therefore:

> **Proceed with database construction only in modified form: a globally extensible, metadata-first, multi-clock evidence registry with three deep cases, parallel affect indicators, explicit missingness, and a rights-aware heritage layer. Do not construct a universal Global Fear Index or identify one global origin year.**

The appropriate historical starting point depends on the claim:

Research claim	Appropriate starting point
Atmospheric heat conceptualisation	1824
Quantitative CO_2–temperature calculation	1896
Explicit fossil-fuel CO_2 and observed warming proposition	1938
Reconstructed global surface-temperature backbone	1850, or 1880 for GISTEMP-based work
High-resolution global reanalysis context	1940
Durable international assessment institution	1988
Treaty-level international recognition	1992
US public-media breakthrough	1988

Research claim	Appropriate starting point
Defensible repeated US direct-worry series	1989
Regional European direct-affect evidence	Provisionally 1988, pending primary questionnaire verification
Harmonised multinational environmental-risk comparison	1993
Broad international climate-seriousness comparison	2000
Near-global repeated climate-threat comparison	2019
Active museum collecting of contemporary climate movements	2019–2020
Formal international recognition of meteorological documentary heritage	2025
Analytically useful thesis boundary	1988–2026, with 1824–1987 as archive prehistory
Most defensible quantitatively comparative period	1993–2020 for selected ISSP items; 2019 onward for near-global threat data; survey-family-specific windows for direct affect

There is therefore no single global beginning of "fear of temperature." There is a sequence of partially overlapping beginnings: conceptual, instrumental, institutional, representational, affective, comparative, and heritage-making. The thesis is strongest when it explains why those clocks do not agree.

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